Code No: 117FZ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, May/June - 2019
OPERATIONS RESEARCH
(Common to ME, CSE, MCT, AME, MIE, MSNT, AGE)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Write the phases of operations research. [2]
b) What are the artificial variables? State their role in solving the LPP model. [3]
c) Explain North-West corner method for finding initial BFS. [2]
d) State the travelling salesman problem [3]
e) State the conditions under which 3 machine n jobs sequencing problem is equivalent 2 machine and n jobs sequencing problem. [2]
f) Derive the optimal replacement policy when time value of money is considered. [3]
g) Write the expressions for determining Mixed strategy for 2×2 games without saddle point. [2]
h) Define i) set up cost ii) holding cost. [3]
i) Write expression for average number of customers present in the basic queuing model and define associated terms. [2]
j) State the Bellman's principle of optimality. [3]

PART-B

(50 Marks)

2.a) Define model. Classify models according to the purpose for which they are developed.
b) A company produces two types of hats. Each hat of first type requires twice as much as labour time as second type. If all hats are of the second type only, the company can produce a total of 500 hats a day. The market limits daily sales of the first and second type to150 and 250 hats. Assuming that the profits per hat are Rs.8 for type A and Rs.5 for type B, formulate the problem as linear model in order to determine the number of hats to be produced of each type so as to maximize the profit and solve it by graphical method. [5+5]

OR

3. Solve the following LPP problem by two-phase method
Max = $2x_1 + 3x_2 + 5x_3$
Subject to $3x_1 + 10x_2 + 5x_3 \leq 15$
$33x_1 - 10x_2 + 9x_3 \leq 33$
$x_1 + 2x_2 + 3x_3 \geq 4$ $x_1, x_2, x_3 > 0$ [10]

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4. A company has three factories I,II,III and four warehouses 1, 2, 3, 4. The transportation cost (in Rs.) per unit from each factory to each warehouse is given in table. The requirements of each warehouse and the capacity of each factory are given below.

	1	2	3	4	Availability
I	25	17	25	14	400
II	15	10	18	24	600
III	16	20	8	13	600
Requirement	300	300	500	500	

Find the minimum cost of transportation schedule with initial BFS is through Vogels method. [10]

OR

5. A company has a team of four salesmen and there are four districts where the company wants to start its business. After taking into account the capabilities of salesmen and the nature of districts, the company estimates that the profit per day in rupees for each salesman in each district is as below.

16	10	14	11
14	11	15	15
15	15	13	12
13	12	14	15

Find the assignment of salesmen to various districts which will yield maximum profit. [10]

6. Two jobs are to be processed on four machines A, B, C and D. The technological order for these jobs on machines is as follows:

Job 1	A	B	C	D
Job 2	D	B	A	C

Processing times are given in the following table:

	Machines			
	A	B	C	D
Job 1	4	6	7	3
Job 2	4	7	5	8

Solve it by graphical method. [10]

OR

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7. The following failure rates have been observed for a certain type of light bulbs:

End of week	Probability of failure to date
1	0.05
2	0.13
3	0.25
4	0.43
5	0.68
6	0.88
7	0.96
8	1.00

The cost of replacing an individual failed bulb is Rs.1.25. The decision is made to replace all bulbs simultaneously at fixed intervals and also to replace individual bulbs as they fall in service. If the cost of group replacement is 30 paise per bulb, what is the best interval between group replacement? At what group replacement price per bulb would a policy of strictly individual replacement become preferable to the adopted policy. [10]

8. Solve the following game by using the principle of dominance: [10]

4	2	0	2	1	1
4	3	1	3	2	2
4	3	7	-5	1	2
4	3	4	-1	2	2
4	3	3	-2	2	2

OR

9. Annual demand for an item is 5400 units. Ordering cost is Rs.600 per order. Inventory carrying cost is 30% of the purchase price/unit/year. The price breaks are shown as:

Quantity	Price(Rs.)
$0 \le Q_1 < 2400$	12
$2400 \le Q_2 < 3000$	10
$3000 \le Q_3$	08

Find the optimal order size. If the order cost is changed to Rs.200 per order, find the optimal order size. [10]

10.a) Explain the Kendal notations for representing the queuing model.

b) A television repairmen finds that the time spent on his jobs has an exponential distribution with a mean of 30 minutes. If he repairs the sets in the order in which in which they came in and if the arrival of sets follows a Poisson distribution with an approximate average rate of 10 per 8-hour day, what is the repairmen‘s expected idle time each day? How many jobs are ahead of the average set just brought in? [5+5]

OR

11. Minimize $y_1^2 + y_2^2 + y_3^2$ subjected to $y_1 + y_2 + y_3 = 15$ using bell. [10]

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